



PROJECT PROPOSAL

‘MINDSCAPE’

MOOD BOARD-BASED PSYCHOLOGICAL SUPPORT SYSTEM FOR STUDENTS

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ABSTRACT

MINDSCAPE is an innovative web-based platform designed to provide mental health support for students through an AI-powered chatbot and community-based discussion forums. The platform combines advanced natural language processing with a supportive community environment to address mental health concerns, offering both immediate AI assistance and peer support through moderated discussion boards.

1. INTRODUCTION

MINDSCAPE is a comprehensive mental health support platform that leverages artificial intelligence and community engagement to provide mental health assistance to students. The platform combines an AI-powered chatbot with moderated discussion forums to create a safe and supportive environment for mental health discussions.

1.1. PROBLEM STATEMENT

Students often face significant mental health challenges but may hesitate to seek help due to stigma, accessibility issues, or lack of immediate support. Traditional mental health resources may not be readily available or may not provide the immediate response that students need.

1.2. PRODUCT OVERVIEW

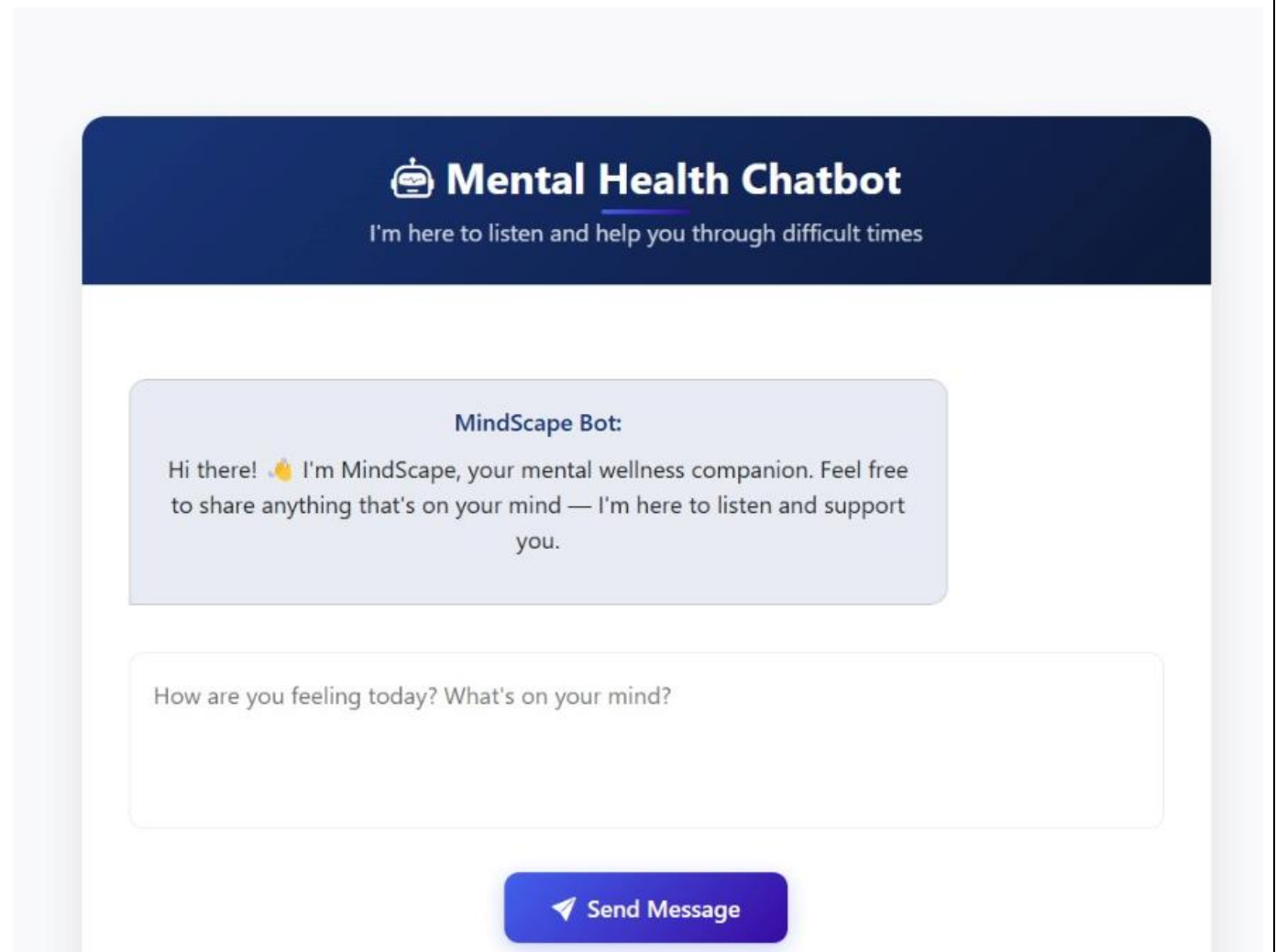
MINDSCAPE addresses these challenges by providing:

1. An AI-powered mental health chatbot for immediate support
2. Moderated community forums for peer support
3. User authentication and profile management
4. Content moderation system
5. Secure and private communication channels

1.3. PRODUCT FEATURES AND SCOPE

AI Mental Health Chatbot

- Immediate responses to mental health queries
- Knowledge base built from reliable mental health resources
- Natural language understanding and response generation



Community Forums

- Topic-based discussion boards
- Moderated content
- Threaded comments
- User reporting system

Your Communities

Connect with like-minded individuals and share your journey in these supportive spaces



Depression

 Created Jan 01, 2024

Join this community to connect with others and share experiences

[View](#) [Leave](#)

 Delete



Stress

 Created Jan 01, 2024

Join this community to connect with others and share experiences

[View](#) [Leave](#)

 Delete



Anxiety

 Created Feb 15, 2024

Join this community to connect with others and share experiences

[View](#) [Leave](#)

 Delete

User Management

- Secure authentication
- User profiles
- Moderator roles and permissions

Ahmad

Email

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Log In

2. STATE OF THE ART

2.1 COMPARISON WITH SIMILAR PRODUCTS

2.1.1. 7 Cups vs MINDSCAPE

- 7 Cups: Focuses on human listeners and paid therapy
- MindScape: Combines AI support with community engagement
- Advantage: MindScape provides immediate AI support while maintaining community aspects

2.1.2. BetterHelp vs MINDSCAPE

- BetterHelp: Professional therapy-focused platform
- MindS: Student-focused with AI and peer support
- Advantage: MindScape offers free, immediate support through AI and community

3. FUNCTIONAL REQUIREMENTS

3.1. AI CHATBOT SYSTEM

3.1.1. NATURAL LANGUAGE PROCESSING

- Uses Ollama's Qwen 0.6B
- Implements RAG (Retrieval Augmented Generation)
- Processes user queries in real-time

3.1.2. KNOWLEDGE BASE

- Web-based resources integration
- PDF document support
- Vector-based storage using FAISS
- Regular updates and maintenance

3.2. COMMUNITY SYSTEM

3.2.1. USER MANAGEMENT

- Registration and authentication

- Profile management
- Role-based access control

3.2.2. Discussion Boards

- Topic-based organization
- Threaded comments
- Moderation tools
- Content reporting system

3.3. MODERATION SYSTEM

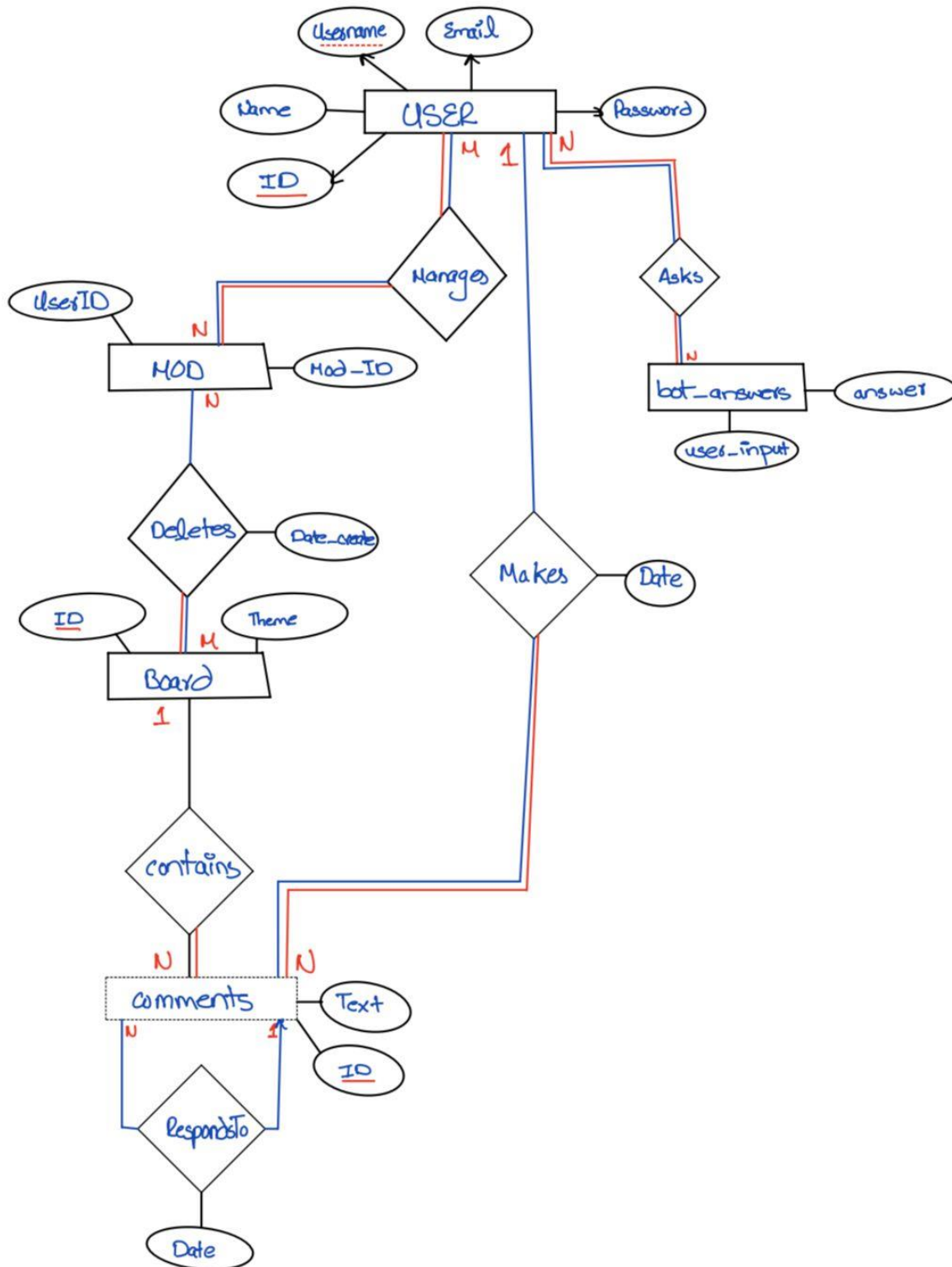
- Moderator appointment
- Content approval workflow
- User reporting handling
- Board management

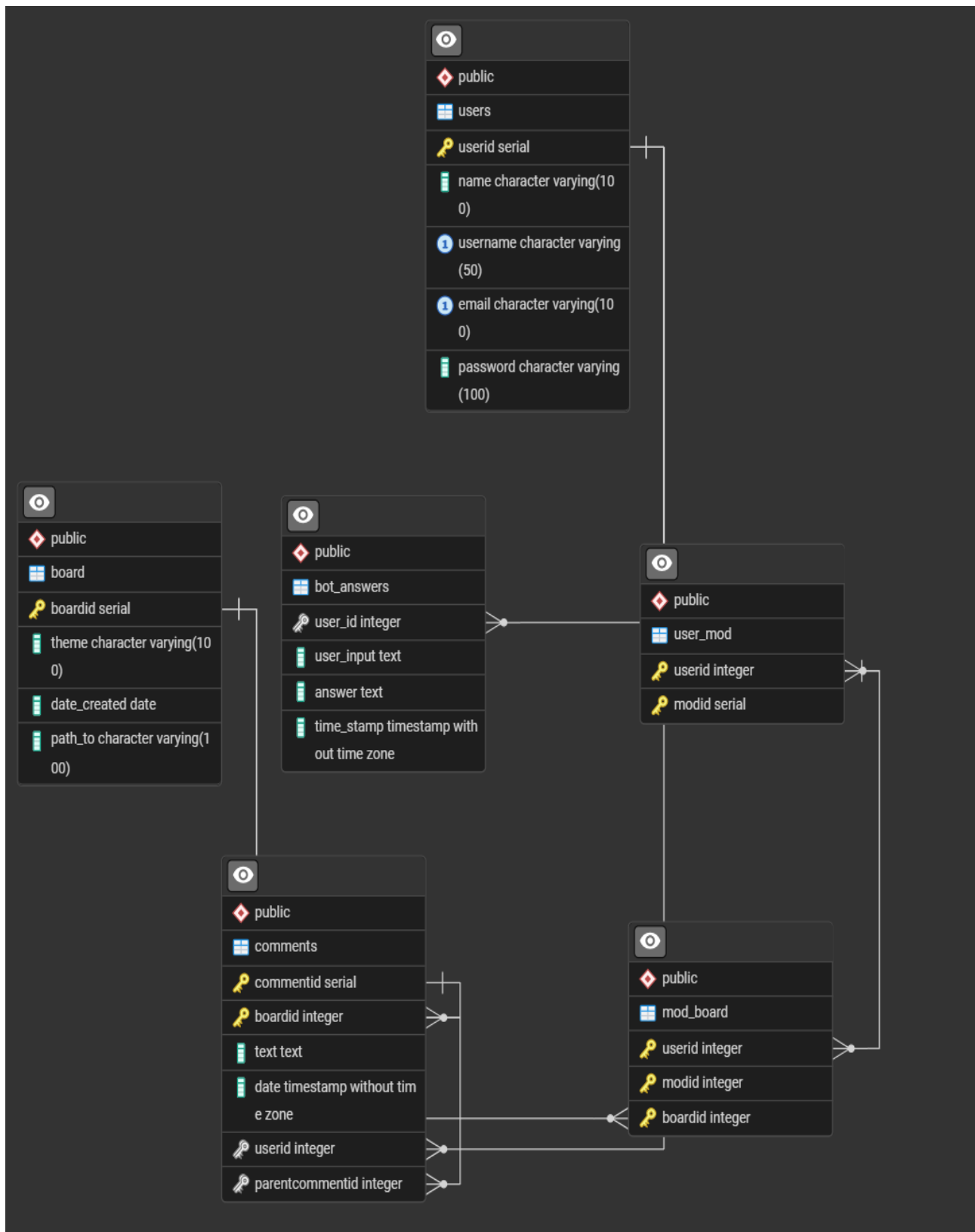
4. ER MODELLING

The database scheme consists of the following entities and their attributes:

ENTITIES	ATTRIBUTES
1. Users	○ UserID (Primary Key) ○ Name ○ Username ○ Email ○ Password
2. User_Mod	○ UserID (Foreign Key) ○ ModID (Primary Key) ○ Links users to moderator roles
3. Board	○ BoardID (Primary Key) ○ Theme ○ Date_Created ○ path_to
4. Mod_Board	○ UserID (Foreign Key) ○ ModID (Foreign Key) ○ BoardID (Foreign Key)

	○ Links moderators to boards
5. Comments	○ CommentID (Primary Key) ○ BoardID (Foreign Key) ○ Text ○ Date ○ UserID (Foreign Key) ○ ParentCommentID (Self-referential)
6. Reports	○ ReportID (Primary Key) ○ UserID (Foreign Key) ○ Text ○ IO (Boolean)





5. UI/UX OF PROJECT

The platform features a clean, intuitive interface with:

1. AUTHENTICATION PAGES

- Login form
- Registration form
- Secure session management

2. CHATBOT INTERFACE

- Real-time chat window
- Message history
- Clear response formatting

3. COMMUNITY INTERFACE

- Board listing
- Threaded discussions
- Moderation controls
- User reporting interface

6. CONCLUSION AND FUTURE WORK

CURRENT ACHIEVEMENTS

- Implemented AI-powered mental health support
- Created moderated community platform
- Established secure user management
- Developed content moderation system

FUTURE WORK

- Improved response accuracy
- Multi-language support
- Emotion detection

COMMUNITY FEATURES

- Real-time notifications

- Private messaging
- Group therapy sessions

ANALYTICS AND MONITORING

- Usage statistics
- Mental health trend analysis
- User feedback system

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